Software Metrics Classification for Agile Scrum Process: A Literature Review

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***Abstract*—Software metrics play an important role to measure every attributes and project success in software development context. In the scrum process, there are several events performed on each sprint in order to monitor team performance. Conducting an objective performance measurement in the agile process meet some challenges due to the differences in team preferences and project complexity. This paper discusses the software measurement metrics and their fundamental role in scrum software development life cycle. We review software metrics on scrum context based on 13 selected literatures. Software metrics are grouped by measurement objectives and scrum events goal. We discuss 34 software metrics distributed in scrum events; sprint planning, daily scrum meetings, sprint review and sprint retrospective. Then, we perform a unique classification and provide a set of recommendation software metrics to implement scrum based on entire project management perspectives to conduct a proper performance measurement.**

Keywords*—*Agile development process; project management; software metrics; scrum development process; scrum framework

I. Introduction

Agile software development method is proposed to overcome the inability of a traditional method to response the changing needs. This is in line with the rapid development of business software field. There are many kinds of agile-based software development method such as Scrum, eXtreme Programming, Kanban, Feature Driven Development, Lean Software Development, and Crystal. Scrum is the most popular framework for agile software development process. Scrum has been adopted in total 73% by software development organization [1]. In the development of agile-based software, each team member works independently and initiatively. Agile practice refers to the principles described in the agile manifesto [2].

Measuring performance and project progress is an interesting field in software engineering studies. There are some challenges in measuring the performance of the agile-based development team [3]. Various software metrics have been proposed in the previous study, which focuses on measuring the development performance from a specific dimension. Performance of software developer affects the quality and the productivity of software development [4][5]. Various types of measurements are also conducted in order to understand the occurred problems and to propose the solutions for the problems.

In the traditional software development process, software metrics are grouped into three types: product metrics, process and project metrics [6]. Oza and Korkala [7] classified the software metrics based on their use, e.g.: code level, productivity/effort level, and economic metrics. In contrast to previous research, we reviewed some literatures that proposed software metrics on the scrum process. We integrated the software metric into the scrum event stages. The result of this research is to classify the software metrics based on their use in the scrum events, then we give a set of recommendation metrics in term of project monitoring based on four perspectives of measurement. This paper consists of five sections that describe the literature review, research methods, results, discussion of the performance measurement metrics and monitoring the progress of the scrum process. In the end, we stated the conclusion of this study.

II. Literature Review

A. Scrum Theory

Scrum is a framework for management process that focuses on developing high-quality software in a short time [8]. Basically, the concept of scrum is also applied in non-IT project management, such as marketing, and in the learning context [9]. The main concept of scrum is a sprint, and each sprint is performed within two to four weeks. The scrum implementation involves a team that consists of the project owner, scrum master, and a development team in total 7 ± 2 members. Each member of the Scrum team has different responsibilities. The characteristics of the scrum team are self-organizing and cross-functional.

TABLE I. Scrum Attributes

Term	Description
Project Owner	The owner of the project contract, he also specifies the software specification and manages the product backlog. Project owner has the right to approve or reject the team work.
Scrum Master	A developer who controls and ensures team members to follow the scrum rules in practice.
Team member	A development team that implements scrum rules (cross-functional).
User story	Description of the client expected features (initial user requirement).
Sprint	Cycle/iteration (time-box based) between 2 to 4 weeks.

Product Backlog	A list of fixed features summarized based user story.
Story point	The abstract measurement of effort required to implement the user story. It reflects the feature complexity.

The software development process using Scrum has been set up in The Scrum Guide [10]. Schwaber and Sutherland explain that scrum consists of four events to create a regular rhythm, thus minimizing the unnecessary meetings. Each event has a purpose and fixed time frame.

1. Sprint Planning

Sprint planning is a step that involves all members of the project (project owner, scrum master, and development team) to discuss the product backlog items to be undertaken into the sprint. Effective planning needs an accurate and reliable estimated. Sprint planning is conducted based on features. The team estimates the effort needed to develop the product backlog using the story point and generates a sprint backlog.

The discussion of sprint planning is divided into two main topics as follows:

a. Review the priority activities on the product backlog list. Product owner explains the purpose and what is desired to the product to be developed (*sprint goal*).
b. Describes the planning activities on sprint and the team choose the product backlog items (PBI) that will be done in the sprint and committed to the discussed terms in sprint planning.

2. Daily Scrum Meeting / Stand Up Meeting

Daily scrum meeting is the main access to review the adaptation process in Scrum practice. Daily scrum meeting is ideally done for 15 minutes in the beginning and is attended by team members who are involved along the sprint. In this brief meeting, each team member states their work and their constraint. They also summarize the targets to be achieved at the next meeting. Scrum master acts as a facilitator for the project to run as expected. At the end of the meeting, the completion time is updated to find out the amount of work to be done during the sprint [11].

3. Sprint Review

The Sprint review is the stage of demonstrating the team's results during the sprint in the presence of all stakeholders (including stakeholders outside of the scrum team) at the end of the sprint. Thus, the customer or the client can see the progress of the project and provide some feedbacks to the released feature. Client response is a base for the team for the next sprint planning.

4. Sprint Retrospective

This discussion aims to introspect the ongoing sprint. The main focus is to improve the communication patterns between team to be an effective collaboration to complete the next sprint. Sprint retrospective is conducted after the sprint review and before the next sprint planning.

B. *Software Measurement Goal*

Software metrics play an important role in measuring every attribute and project success of the software development. The quality and productivity of the team should be measured accurately. Qualitative and quantitative measurement metrics are proposed to figure out and to analyze various development processes undertaken, improving the quality and productivity of the team. The use of metrics will change the team behavior in order to follow the defined rules set by the company.

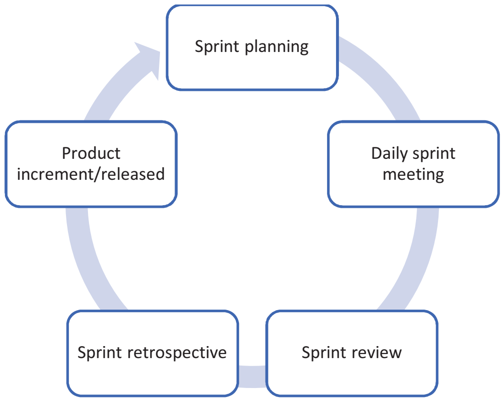

Figure 1. Scrum event flow

Software metrics can improve the development process, produce a better project and control the quality, and improve estimation process [12]. Software metrics can be used to provide an objective view and to evaluate the various impacts of software development process changes. In addition, software metrics also provide benefits to review project progress, to monitor the product quality, and to forecast the project and the project management [13].

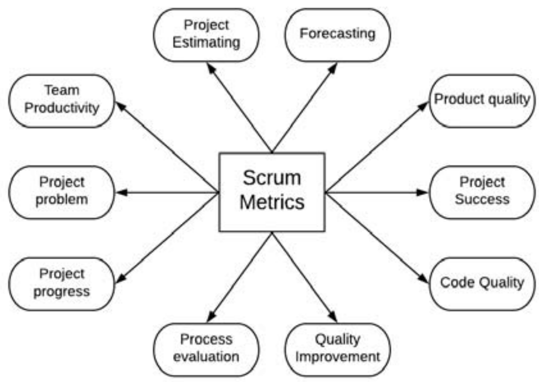

Figure 2. Software measurement goal

In the initiation process, software metric is used to determine the project scope, to determine the business priority features, and to make estimates for the next sprint [14]. In agile concepts, communication and collaboration are the basic aspects to be improved. Kupiainen et al., [15] concludes that software metrics are used for specific purposes, such as for project planning and controlling the running of sprints, reviewing project progress, understanding and improving quality, finding solutions to solve problems and motivating teams. In general, software metrics used to track the team members' involvement intensity, improve communication within the team and measure the software quality to describe project success.

We conclude about the scrum event based on The Scrum Guide [10] and the measurement objectives in software development based on the literatures shown in Table II.

TABLE II. SCRUM EVENT ACTIVITIES

Scrum event	Attendance	Meeting goal	Validation
Sprint planning	Project owner, scrum master, and development team.	Project scope	Attendance agreement
		Feature prioritization	
		Effort Estimation	
Daily scrum meeting	Scrum master and development team.	Measure the goal achievement	Backlog progress
		Defines the constraints	
Sprint review	The project owner, Scrum master, development team and client.	Deliver value	Working software
		Demonstrate the features	Client feedback
Sprint retrospective	Project owner, Scrum master, and development team	Figure out and motivate teams	Team collaboration and process improvement in the next sprint
		Communication improvement	

III. RESEARCH METHOD

The research method used is literature study on software measurement model of development in scrum process framework. The authors studied various metrics discussed in 13 research papers, both journals and conferences. We consider some literatures published from 2007 to 2017 use this string search (“Scrum” AND “Metric” AND “Performance”). We applied the string search through some digital library, such as: IEEE Xplore, Science Direct and ACM. We include the paper based on the criteria below:

- The papers give an explanation about scrum methodology and its implementation.
- Discuss the software metrics to measure the team performance.
- Give the guidelines to use the software metrics.

The result of this research is the classification of software metrics based on their use of scrum events: sprint planning, daily scrum meeting, sprint review and sprint retrospective. Software metrics are classified based on the general purpose of using metrics and the goals of each scrum event. Then we tried to propose a set of software metrics to measure the team performance based on some perspectives.

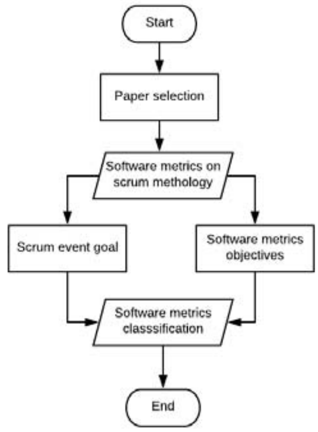

Figure 3. Research method

IV. SOFTWARE METRICS CLASSIFICATION

In total 34 Software metrics discussed in the primary studies are grouped into four groups, they are: sprint planning, daily sprint meeting, sprint review and sprint retrospective. The goal of software metrics grouped are to monitor the team performance and project progress on each scrum event. The classification for each scrum event can be found in Appendix. Appendix 1 consists of 5 common metrics used when team conduct the sprint planning. In appendix 2, we collect 6 metrics used to measure the team progress in the daily sprint meeting. Table for Appendix 3 and 4 consist of metrics used in the sprint review and sprint retrospective meeting. We discuss these metrics in the next chapter.

V. FINDINGS AND DISCUSSION

Scrum software development method is characterized by a short development cycle, frequent deliveries, frequently meeting and learning. We provide a unique classification for each process to recommend a set of appropriate metrics. Based on the tables of classification above, we recommend a set of software metric based on the frequently used and successful metrics based on literatures. We highlight two most influential metrics for each scrum event from some perspectives.

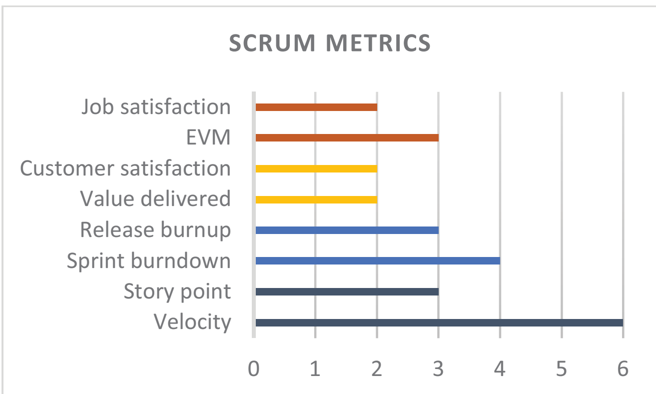

Figure 4. Frequently discussed metrics based on literatures

Sprint is planned based on the selected item on the product backlog list, estimate effort needed to complete the task to achieve the sprint goal. Usman et al. [16]found estimation plays a pivotal role during the sprint and release planning for the agile team. The two most popular metrics in the sprint planning stage are story point and velocity. We found 3 literatures discuss the use of story point [17][18][19] and 6 literatures for velocity [19][3][18][17][20][13]. Scrum team conduct 'planning poker' to determine the story point for each product backlog [16]. The team productivity when the sprint in progress is calculated by velocity. Then, velocity is used as reference to predict the amount of work that can accomplish in the next sprint and estimate the number of sprint required to complete the project. The output of sprint planning is a fixed time of deliver value.

Daily stand up meeting goal is to track the team activity and to make sure team member using the time efficiently by intensive reporting. The metrics to measure this scrum process are sprint burndown and released burndown chart [18][20][21][22]. The Burndown chart (sprint and release) is used to show the project progress based on the total work and duration of the remaining sprints. It indicates the team performance during the development process. In addition, the burndown chart provides an overview of the project trend that can be used as a reference to determine the project completion time. The goal of these metrics is to monitor the whole project and team activities.

In the sprint review process, scrum team conduct a meeting to demo and to validate the completed features with the stakeholders, especially the client. Some researchers measure the successful of the sprints by value delivered [20] and customer satisfaction [23][24]. Business value is used to conclude the release. For example, 80% of desired product backlog done may decide that it is enough to call as a release. While customer satisfaction might be a step to confirm and to justify between project planning and the result.

TABLE III. SOFTWARE METRIC PERSPECTIVES

Software Metrics	Team	Project	Management	Client
Velocity	√			
Story point		√		
Sprint burndown		√		
Release burnup		√		
Value delivered	√			
Customer satisfaction	√			√
EVM		√	√	
Job satisfaction	√		√	

Sprint retrospective aims to assess the recent sprint and plan a possible improvement for the next sprint. A major focus of software metrics in sprint retrospective is to inspect the team performance itself and motivate team how to build team engagement in all aspect leading the organization and project goal. Measure the EVM [18][25] and the job satisfaction of development team [24] have a high impact to analyze the successful of recently sprint. Measuring the team satisfaction related to human aspect has a bigger effect on software developer productivity [26]. It gives organization the clear understanding about aspect that need to be improved. Earn business value (EVM) is not part of scrum but required as a good practice of measurement. Further explanation regarding EVM in agile is discussed in this study [27]. By measuring the SPI and CPI in daily basis, thus enabling clear response for any deviation from project planning (schedule and cost perspective).

VI. CONCLUSION

A proper performance measurement in term of software project involve several levels of perspective. In this research, we provide a set of software metrics to measure the scrum team performance on each event based on four perspectives of measurement. We conclude this article as follows:

- In the sprint planning stage, we recommend use the story point and velocity as the basis to create a planning. While velocity is used to measure the team reliability, story point represents the project size and the backlog complexity.
- Sprint burndown and release burndown is used to continuously monitor the team activities. They represent the number of work remaining in sprint and project in story point. Both of this chart is related to project progress.
- Sprint review is the most valuable event because the team need to validate their work, whether it meets the client expectation or not. To review the features to be released, business value delivered and customer satisfaction is the main indicator of measurement in the sprint review.
- The sprint retrospective is conduct to analyze the recent sprint, the EVM and job satisfaction of the development team is used as the indicator of this measurement. EVM goal is to measure the success of project from the management perspective and conduct a job satisfaction survey of the team is a learning basis for organization to perform better in the next sprint.

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APPENDIX

APPENDIX 1. SPRINT PLANNING METRICS

Software metrics	Measurement Process	Description
Effort estimate [28]	Determining the story point per user story to plan the sprint.	Aims to estimate the effort required per sprint.
Story point [17]	Describing the difficulty level of a story using Fibonacci format. Difficulty could be related to complexities, risks, and efforts involved.	The measurement process to predict the software size and effort required.
Task effort [14]	Team generates and estimate effort for each task based on programmer hours.	The measurement unit used to estimate the effort required to implement the user story.
Task's expected and end date [28]	These dates are determined after considering the priority of a task in a sprint.	Fixed time to complete the task.
Velocity [3][18] [17][13][20][19]	∑ of all accepted work.	The number of work complete during sprint.

APPENDIX 2. DAILY SPRINT METRICS

Software metrics	Measurement Process	Description
# of an open defect [23]	A number of open defects/days calculated at the time of release.	Visualized using a chart showing the open defect for the current release compared to the same relative date in the release cycle before implementing scrum.
Contribution [29]	Count the individual meeting point based on the check-in requirement.	Assess the direct participation and quality of engagement during the daily scrum meeting.
The ratio of work spent and work remaining [24]	Work spent on day *i* for each task *j* in the sprint backlog/Work remaining on day *i* for each task in the sprint backlog.	The ratio between the number of tasks (per day) done and the remaining task (per day).
Standard violation [20]	Track the number of standards violated per sprint. This metric directed the team toward the right behaviors during the development.	The code quality standards that have to be fulfilled by the team.
The release burndown chart [18][20][21][22]	Represent the amount of work remaining and its reduction in each sprint by plotting the sum of story points of all unfinished stories in the Product Backlog.	A graph to describe project progress (ideal vs the actual progress) based on remaining tasks in a timeframe (sprint).
The sprint burndown chart [18][20][21]	Described by total day in a sprint against the number of remaining a working hour.	A chart to show how much work remains in the sprint and whether teams are on schedule.

APPENDIX 3. SPRINT REVIEW METRICS

Software metrics	Measurement Process	Description
# of defects found in system test [23]	A total defect found during system test after reduced by defect removal techniques, such as: test-driven development, pair programming, peer review.	A total defect found in system test to measure the code quality being delivered to the system test process.
Bug correction time from new to the close state [13]	Not clearly defined in the primary study.	Leads to the time it takes to fix the bugs found during the sprint.
Business value delivered [20][23]	Measured in term of story point, number of story, or another abstract	Measure how much value the business attaches to a single feature or story.

Software metrics	Measurement Process	Description
	measurement to describe the business value achieved.	
Customer satisfaction [23][24]	Based on the product feature demo.	Qualitative assessment of customer satisfaction based on certain indicators.
Completed web pages [28]	The number of web pages that have been completed.	The way team reported their progress by comparing the work-planned and work-completed.
Defects deferred [23]	Measured by the number of defects discovered by a customer in the field.	A number of defects encountered, but not fixed upon release.
Defect per iteration [20]	Calculated all the defect that was introduced during the sprint.	The defect that was found and should not be a pile up in the sprint.
Delivery on time [13]	The ratio of features done in the planned release schedule.	This is indicated the real customer value by delivering the targeted features.
Error density [24]	- Number of error found in a sprint - Number of error reported by the user after release (calculated per sprint) - Total lines of code in a sprint.	The error density is total found error by computing sums over all PBIs in the sprint or release.
Focus factor [3]	Velocity ÷ Work capacity.	The ratio between velocity and Work capacity, the good value ranges around 80% for a team.
Fulfillment of scope [24]	The ratio between the number of PBIs actually implemented and the number of PBIs committed.	Indicated the team to fulfill the agreed commitment to the sprint planning.
Number of stories [20]	This metric is calculated as a simple count or weight by story complexity, such as simple, medium and complex of the number of stories in the sprint.	Tracking the project progress based on the number of stories accepted.
Open defect severity index [13]	Measured at the end of the sprint. This is avoided by the team, if they had any, they will bring it to the retrospective and discuss to avoid it in the future.	Measure the quality of feature in each iteration.

Software metrics	Measurement Process	Description
Percentage of Adopted work [3]	Σ Original Estimates of Adopted work / Estimation forecast for the sprint.	Adopted work is the new work added from product backlog to the sprint because team has already done the sprint backlog.
Percentage of Found work [3]	Σ (Original Estimates Found Work) / (Estimation Forecast for sprint).	Found work is the unexpected work added to the sprint to complete the sprint backlog.
Progress chart (Scrum board) [20]	Shows the project progress by task status (not started, in progress and complete) on the board.	A tool to track the progress of team member.
Unit test coverage for developed code [13]	The measurement shows the percentage of unit test cases coverage for a code in a product and is reported at the end of each sprint.	Relates to the testing between expected function and the results.
Work capacity [13]	The value of work capacity should be equal or more than velocity.	The number of tasks reported during the sprint, regarding the completed feature or not.

APPENDIX 4. SPRINT RETROSPECTIVE METRICS

Software metrics	Measurement Process	Description
Earn value management (EVM) [18][25]	Calculated the Schedule Performance Index and Cost Performance Index required.	A technique for measuring project performance and progress in an objective manner
Impression [29]	Conducting a survey with a questionnaire that will be answered by the team involved	Review team members' work based on the other team members opinion.
Influence [29]	Average (individual events per day) ÷ Average (total team events per day).	Measures individual involved and participation in project progress.
Job satisfaction [24][26]	Conducting a survey using questionnaire related to the work done.	Developer's personal satisfaction with his job.
Net promoter score [23]	Measure the customer satisfaction and feedback using qualitative survey.	Measure the client satisfaction and its impact. Satisfaction leads the customer to recommend products to other potential customers.